

Diagnosing Standardization Gaps in Nutrition Evidence: The Nine-Layer Framework and the Physics Control Case

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Erratum, v1.4 (31 August 2026) — Layer 2 Identity, nutrition column

Versions up to v1.3 printed 33 for Layer 2 Identity in the Stack Audit recalibration of 15 August 2026, and 70 in Table 1 as published on 23 August 2026. The two numbers were produced from the same evidence and the same stated deficiency. No derivation for 70 was recorded, and none has been found. Table 1 is corrected to **33**.

The recalibration derived 33 in writing — “Layer 2 — Identity. Estimated 35%. Verdict: defensible → 33%” — on the grounds that packaged-food identity is solved by GS1’s GTIN while generic-food identity and cross-registry resolution are not. The justification printed beside 70 stated that same deficiency, so the correction restores consistency between the score and its own stated reason rather than introducing new evidence.

The move was two rubric bands — from the 30–49 band to the 70–89 band, skipping 50–69 — on one of only two layers whose name did not change in the 23 August taxonomy migration, so no concept change accounts for it.

Root cause. Both readings were defensible against the §2 rubric as written, because the rubric did not say whether a score measures that mature standards *exist* or that the field *has and uses* one. That underspecification is the defect; the wrong number was its symptom. This version adds an explicit construct definition (§2.2), and Layer 2 was rescored against it rather than simply reverted. Contributing cause: Table 1’s basis line for Layer 2 cited vocabulary artifacts (FoodOn, LanguaL, FoodEx2, INFOODS tagnames), which are Layer 3 evidence — Layer 2 was in part scored on the strength of the layer above it. The v1.4 rationale argues from identifier coverage and cross-registry resolution instead.

Consequence. With Layer 2 at 70, exactly one nutrition layer met the 60-point layer threshold of §5. At 33, none does, and the highest score in the nutrition column is 55 at Layer 1. The v1.3 text describing a profile that “peaks at Identity” is corrected throughout; see §4 and §5.

Citation correction. Versions up to v1.3 printed the version DOI of the first deposit (10.5281/zenodo.22070407) as the paper’s citation in the header and page footers. That record is superseded and is not where v1.3 was hosted. The concept DOI 10.5281/zenodo.22070406 should be used for all citation.

Abstract

Nutrition science is frequently criticized as “unstandardized,” but the criticism conflates two distinct problems: the irreducible biological heterogeneity of food and diet (a property of the object of study) and the standardization of the formats in which measurements, records, and judgments are made (a property of the field’s infrastructure). This paper introduces the Nine-Layer Stack, a diagnostic taxonomy that separates format standardization into ten addressable layers (L0–L9) across four strata — Measurement, Record, Judgment, and a cross-cutting Traceability layer. We argue that standardization compounds downward: a field’s effective standardization is set by its weakest format layer, not its average. We use physics as a calibration control: it attains near-total standardization at the measurand, identity, vocabulary, units-governance, and traceability layers, with no mandatory reporting checklist analogous to CONSORT and no research statute. It manages this because scope travels with the claim — every physical law is stated with its regime of validity attached. Applying the framework to nutrition, we find that no format layer meets the 60-point layer threshold: the field’s highest score is 55, at Instrument/Method, against physics, which meets it at all ten layers. Nutrition therefore fails the first of the two conditions in our functional floor, and fails it at every layer. Identity is the sharpest case — rich food ontologies and reference databases exist, yet the layer scores low because they do not resolve the field’s basic unit, a generic unbranded food — and it is the stack’s clearest demonstration that the existence of standards is not standardization. The collapse is deepest at Scope and Translation, and the stack is capped by a Layer-0 ceiling: many nutritional “measurands” (e.g., “a tomato”) are categories, not well-defined quantities. We propose a keystone intervention: a machine-readable StudyRecord governed by a minimum-information checklist (“MI-Nutrition”) that would move the study-record, translation, and traceability layers together — three of the four lowest-scoring layers — following the community-adoption path of MIAME rather than a statutory mandate. Layer scores are presented as calibrated ordinal expert judgments, not measurements. Novel terms (functional floor, population fence, value lane, MI-Nutrition) are proposals of this paper.

Keywords: nutrition science; standardization; metrology; food composition databases; reporting guidelines; evidence synthesis; minimum information standards; research reproducibility; ontology; metrological traceability

1. Introduction: the false binary

A recurring claim in both scholarly and popular commentary is that nutrition science is not “real science” because its findings are unstable: effect estimates shift between studies, dietary guidelines appear to reverse, and observational associations often fail to replicate. This critique typically bundles two problems that must be kept separate.

The first is **biological heterogeneity**. A tomato grown in one soil in one year differs measurably from a tomato of the same cultivar grown elsewhere; two people eating the identical meal absorb different fractions of its energy and micronutrients. This variation is a property of the object of study, not a defect in method. Physics has irreducible variation too — the point of an uncertainty budget is precisely to quantify it — but no physicist calls the spread of repeated measurements a “standardization failure.” Variation in the sample is a feature of the world.

The second is **format standardization**: whether the field has agreed, stable formats for defining what is measured, recording how it was measured, and reasoning about what the measurements mean. This is infrastructure, and it is separable from heterogeneity. A field can face enormous biological heterogeneity and still be highly standardized in format (clinical laboratory medicine is an example), or it can study relatively stable objects and be poorly standardized in format.

The confusion between these two problems produces two errors. Optimists point to heterogeneity as “just biology” and conclude nothing is wrong with the infrastructure. Pessimists point to unstable findings and conclude the science is worthless. Both are wrong because they are measuring the wrong thing. This paper offers a taxonomy to separate the two, so that the genuinely fixable problems — format problems — can be diagnosed layer by layer.

That a field of this consequence might simply never have been standardized is an observation made from outside it. Thiel and Masters, asking which fields “matter but haven’t been standardized and institutionalized,” nominate nutrition as their example: a subject that matters to everybody, that cannot be taken as an undergraduate major at Harvard, and that consequently loses most of its top scientific talent to other fields — with the result, as they put it, that far more is known about the physics of distant stars than about human nutrition.¹ The observation is cultural rather than empirical, and this paper does not rest on it. But the coincidence of examples is worth noting, because the pairing that a venture capitalist reached for casually is the pairing this paper adopts deliberately: physics as the calibration control, nutrition as the case.

The field’s own literature bears the observation out, at least as a floor. Fewer than 0.3% of the nutrition literature engages standardization as a problem: in a 2,022,437-record corpus, 5,509

titles or abstracts mention standardization, harmonization, interoperability, reporting guidelines, minimum-information checklists, controlled vocabularies or FAIR — 27 per 10,000 — while “standardized,” asserting that a study’s own materials were controlled, appears 5.7 times as often.²

The critique is not only external, and not only recent. Penders and colleagues (2017), writing in a nutrition journal, ask whether the field is capable and credible on its own terms and argue that its difficulties are as much institutional as methodological. That is the distinction this paper’s taxonomy is built to make precise: institutions build formats, and formats are what can be audited layer by layer.

We deliberately place one thing out of scope: the education and public-transmission layer (how nutrition science reaches consumers, journalists, and clinicians). Transmission failures are real but downstream of the evidence-production problems we analyze here.

2. The Nine-Layer Stack

We propose a taxonomy of ten layers (numbered L0 through L9) grouped into four strata. The name “Nine-Layer Stack” reflects the nine substantive format layers L0–L8; L9 is a cross-cutting traceability layer that binds the others.

Measurement stratum — L0 — Measurand. Is the thing being measured stable and well-defined? In metrology, a measurand is “the quantity intended to be measured” (VIM, JCGM 200:2012), and its specification requires knowing the kind of quantity and the state of the body or substance carrying it. L0 asks whether the field’s core quantities are genuine measurands or merely categories. **L1 — Instrument/Method.** Are the instruments and analytical methods calibrated, validated, and comparable across laboratories?

Record stratum — L2 — Identity. Is each entity (a food, an organism, a sample) assigned a persistent, unambiguous identifier? **L3 — Vocabulary & Reference Data.** Are there shared controlled vocabularies and authoritative reference datasets? **L4 — Study Record/Provenance.** Is the full provenance of a study — what was done, to what, when, with which database version — captured in a structured, machine-readable record? **L5 — Reporting.** Are results reported against a shared checklist so that readers can appraise and reproduce them?

Judgment stratum — L6 — Synthesis/Grading. Is there a shared, applied system for grading the certainty of a body of evidence? **L7 — Translation/Scope.** Is the scope of a claim — the population, exposure range, and outcome to which it applies — bound to the claim itself, so it can be translated to a new context without silent extrapolation? **L8 — Governance/Incentives.** Are there governing bodies and incentive structures that maintain the standards over time?

Cross-layer — L9 — Traceability. Can a downstream number be traced through an unbroken documented chain back to its origin? Metrological traceability is “the property of a measurement result whereby the result can be related to a reference through a documented unbroken chain of calibrations” (VIM 2.41).

The compounding principle. Standardization compounds downward: an error or ambiguity at a lower layer propagates into and cannot be repaired by higher layers. If the measurand (L0) is ill-defined, then perfect instrument calibration (L1), flawless identifiers (L2), and rigorous grading (L6) still yield a precisely recorded, well-graded number for an ambiguous quantity. Consequently a field’s effective standardization is governed by its weakest format layer, not by its average across layers. A single collapsed layer caps the whole stack. This is why we score fields by their floor, not their mean.

2.1 Scoring method

Throughout, we assign each field an ordinal 0–100 score per layer. **These scores are calibrated expert judgments, not measurements.** They are intended to make relative comparisons legible (physics at L0 is far higher than nutrition at L0), not to assert metric precision. The rubric anchors: ~90–100 = a stable international standard exists and is near-universally used; ~70–89 = a standard exists and is widely but not universally applied; ~50–69 = competing standards exist or adoption is partial; ~30–49 = fragmentary standards, low adoption; ~0–29 = no shared standard. Scores were assigned by the author against the documented infrastructure cited for each layer and revised across successive drafts; they have not been externally validated (see Limitations).

2.2 What a score measures

A score is a single ordinal judgment of **standardisation achieved**, which is existence **and** uptake together, not either alone. The rubric’s top anchor is “a stable international standard in near-universal use,” so use sits inside the construct at every band rather than forming a second axis.

A layer where excellent standards exist and nobody uses them scores **low**, because the layer is not standardised. The maturity of the artifacts is recorded in the per-layer rationale and in the standards catalogue, never as score credit.

The basic-unit rule. Judge the field’s own basic unit of work, not its easiest case. For food identity the basic unit is a generic unbranded food, not a packaged product: solving barcodes while “a raw apple” has no stable identifier is a solved sub-case inside an unsolved layer.

Uptake evidence means an institution requiring, shipping, or gating on the standard. It is **not** corpus mention counts, which this project treats as a floor on use and never as adoption.

This subsection was added in v1.4. It is the fix for the defect recorded in the erratum above: between 15 and 23 August 2026 Layer 2 moved from 33 to 70 on identical evidence, and both readings were defensible against a rubric that had not said which of the two things it measured. The construct is now stated so that the question is decidable.

3. Physics as a control case

Physics is useful here not because nutrition should imitate it — the objects are radically different — but because physics is a *calibration control*: a field that is near-maximally standardized in format, which lets us see what full standardization does and does not require. The instructive finding is what physics standardizes and what it does not.

3.1 What physics standardizes

L0 Measurand and L8 units-governance. Since 20 May 2019, all seven SI base units have been defined in terms of exact numerical values of seven defining constants (the caesium hyperfine frequency $\Delta\nu_{\text{Cs}}$, the speed of light c , the Planck constant h , the elementary charge e , the Boltzmann constant k , the Avogadro constant N_{A} , and the luminous efficacy K_{cd}), as set out in the 9th edition of the SI Brochure published by the Bureau International des Poids et Mesures (BIPM). The numerical values of the seven defining constants have no uncertainty. The governing treaty, the Metre Convention, was signed on 20 May 1875 and, per the BIPM member-states page verified on 23 August 2026, has 68 Member States and 35 Associate States and Economies; membership is still growing — Albania and Peru acceded as Member States in June and August 2026 respectively, 151 years after the treaty was signed. This is the units-and-commerce tier of governance, and it is backed by statute — but note carefully what tier: legal metrology, administered internationally through the International Organization of Legal Metrology (OIML), governs commerce, trade, and engineering, not the conduct of physics research.

L1 Instrument and uncertainty. Physics has a shared grammar for measurement uncertainty — the Guide to the Expression of Uncertainty in Measurement (GUM, JCGM 100:2008) — and laboratory competence is accredited under ISO/IEC 17025:2017. Reference data are curated: the CODATA Task Group on Fundamental Constants issues internationally recommended values of the fundamental physical constants via a least-squares adjustment on a four-year cycle (2014, 2018, 2022), published for public use by NIST.

L2 Identity and L3 Vocabulary. The Particle Data Group’s *Review of Particle Physics* compiles, evaluates, and averages measured particle properties, applying a documented scale-factor procedure to enlarge uncertainties when input measurements are mutually discrepant — an explicit, reproducible rule for combining inconsistent data. Physics uses ISO 80000 (“Quantities and units”) for quantity symbols and the American Physical Society’s Physics Subject Headings (PhySH), introduced in 2016 to replace the older Physics and Astronomy Classification Scheme (PACS, whose final version was PACS 2010).

L4/L9 Provenance and traceability. Preprints are deposited on arXiv (founded 1991 by Paul Ginsparg, now operated by Cornell), the high-energy-physics literature is indexed in INSPIRE-HEP (built and operated by CERN, DESY, Fermilab, and SLAC, with later partners), and underlying scattering data are archived in HEPData (the Durham High Energy Physics Database). In astronomy, the Flexible Image Transport System (FITS), first defined in 1981 and formally endorsed by the International Astronomical Union in 1982, gives a single durable data format. The CIPM Mutual Recognition Arrangement (1999) ties national metrology institutes’ measurements together through an unbroken chain of comparisons.

A convention at the synthesis layer. Particle physics uses the “5-sigma” discovery threshold — a statistical significance of five standard deviations, corresponding to a background-only p -value of about 3×10^{-7} — as the conventional bar for claiming a new particle, as applied in the 2012 discovery of the Higgs boson by ATLAS and CMS. It is a shared convention, not a statute.

3.2 What physics does NOT standardize

Two absences are decisive for the argument of this paper.

First, physics has no mandatory reporting checklist analogous to CONSORT. There is no “physics CONSORT” that a paper must complete before publication. Reporting norms are transmitted through community convention, referee expectation, and the structural discipline of the data repositories (a paper claiming a measured branching ratio is expected to place its data in HEPData), not through a checklist enforced as a condition of publication.

Second, there is no research statute governing how physics experiments must be conducted or reported. Statutes exist only at the units/commerce/engineering tier (OIML legal metrology, national weights-and-measures law). No law dictates how a physicist measures the muon anomaly.

How, then, does physics achieve near-total standardization at the scope layer (L7) without either a checklist or a statute? The answer is that **scope is part of the claim**. Every physical law is stated with its regime of validity attached: Newtonian mechanics is understood to hold for speeds much less than c and actions much greater than $\hbar$; an ideal-gas law is stated with its temperature and pressure range. A physicist does not extrapolate a result outside its stated regime, because the regime is written into the statement of the result. Scope discipline is linguistic and cultural, embedded in how claims are phrased, not procedural. We flag the epistemic status of this claim explicitly, as we do for the reporting-checklist absence (Appendix A, claim 13): “regime of validity” is a disciplinary norm transmitted through training, textbooks, and refereeing, not a citable governing standard, and the high Layer-7 score we assign physics rests on observed practice rather than on a standards document (Appendix A, claim 31). With that caveat stated, this remains the single most important lesson physics offers nutrition, and we return to it in Section 5.

3.3 Table 1. Nine-Layer comparison: physics vs nutrition

Scores are calibrated ordinal expert judgments (0–100), not measurements. See §2.1 scoring method, §2.2 construct, and Limitations. The Layer 2 nutrition score is corrected in this version; see the erratum.

Layer	Physics (author’s est.)	Nutrition (author’s est.)	Basis
L0 Measurand	95	25	Physics: SI constants fix base quantities exactly. Nutrition: “a tomato” is a category, not a measurand; “dietary fibre” is defined by the assay used to measure it.
L1 Instrument/Method	90	55	Physics: GUM + ISO/IEC 17025 + CIPM MRA. Nutrition: validated AOAC methods exist but several coexist per analyte and return systematically different values for the same food.
L2 Identity	90	33	Physics: PDG particle numbering, IAU nomenclature, IUPAC species. Nutrition: packaged identity is solved by GS1’s GTIN, but there is no universal stable identifier for a generic unbranded food, and cross-references between registries (FDC ID ↔ LanguaL ↔ FoodEx2 ↔ FoodOn) are partial and hand-curated. <i>Corrected from 70; see erratum.</i>
L3 Vocabulary/Reference	92	45	Physics: CODATA constants, ISO 80000, PhySH. Nutrition: vocabularies (FoodOn, CDNO) are strong; reference data are not — 17 of 101 food-composition databases satisfy all 13 FAIR criteria evaluated. EN 16104:2012 (CEN/TC 387) exists as a formal standard but is a paid CEN document with compiler-culture rather than universal uptake.
L4 Study Record/Provenance	80	30	Physics: arXiv/INSPIRE/HEPData/FITS. Nutrition: the gap is the study record, not provenance as such — no machine-readable study-record standard carries provenance, and nothing carries the food-value documentation into a record of the study that used it.
L5 Reporting	70	40	Physics: strong convention, no checklist. Nutrition: STROBE-nut exists but uptake is limited; journals endorse it and none gates publication on it.
L6 Synthesis/Grading	75	35	Physics: PDG scale factor, 5σ convention. Nutrition: GRADE exists and is the accepted default; it is applied in a small minority of reviews.
L7 Translation/Scope	90	20	Physics: regime of validity bound to every law; applied translation through engineering codes. Nutrition: scope rarely bound to the effect estimate.
L8 Governance/Incentives	85	40	Physics: BIPM/CGPM/CIPM at the units tier; no statute over research conduct. Nutrition: FAO/INFOODS, journals, agencies — no unifying governance and no research gatekeeper.
L9 Traceability	88	30	Physics: CIPM MRA unbroken chains. Nutrition: imputation status <i>is</i> recorded by the major compilers (EuroFIR, USDA FDC, FAO/INFOODS); the failure is carry-forward, not capture — it is almost never carried into the study, the synthesis or the guideline.

Reading the column. Physics meets the 60-point layer threshold at 10 layers of 10; nutrition meets it at 0 of 10, its highest score being 55 at L1 Instrument/Method. The layer threshold is the first of the two conditions in the §5 functional floor, which is a field-level test and is not scored here. Table 1 is generated from the score register (Appendix D), which is canonical.

3.4 How physics got there: a dated history of the control case

This section reconstructs the actual historical sequence by which physics acquired the layered standardization we score at ~70–95 per layer in Table 1, and then dates each layer’s crossing of the ~60/100 layer threshold. The central empirical finding is that physics did not standardize as a single event. It standardized measurand-first and bottom-up over roughly 125 years, with the units layer leading by more than a century and the reference-data, uncertainty, and traceability layers lagging far behind. Under the paper’s own weakest-layer rule, physics satisfied the layer threshold at every layer not in 1875 (the treaty most people associate with “standardized measurement”) but only in the 1990s, once the uncertainty grammar (GUM, 1993) and the international traceability chain (ISO/IEC 17025 and the CIPM MRA, 1999) were in place.

3.4.1 The arc in brief

Physics standardized measurand-first, crisis-driven, and over more than a century. The metric system emerged from revolutionary France (1791–1799) as the response to a units chaos that

Alder tallies at some 250,000 local variants under roughly 800 names; transatlantic telegraphy forced coherent electrical units (BAAS committee, 1861–1874; CGS 1874); the Metre Convention (1875) created the treaty tier and its institutions (BIPM/CGPM/CIPM); national metrology institutes anchored physical traceability from 1887 (PTR, then NPL 1900 and NBS 1901). Reference-data governance lagged the treaty by half a century; the first systematic least-squares adjustment of the constants was Birge’s 1929 review — literally the first article in the first issue of *Reviews of Modern Physics* — and durable institutional custody arrived only with CODATA (founded 1966; Task Group on Fundamental Constants, 1969; regular four-year cycle from 1998). The literature and synthesis infrastructure grew inside the community rather than by mandate (Rosenfeld particle tables 1957, PDG 1968 and its scale-factor averaging rule; SPIRES 1974; arXiv 1991; the 5 σ discovery convention consolidating culturally by the 1990s). The uncertainty grammar and the international traceability chain closed last — GUM in 1993, ISO/IEC 17025 and the CIPM Mutual Recognition Arrangement in 1999 — and the measurand layer itself was completed only in 2019, when the 26th CGPM fixed every base unit to a defining constant. Table 2 carries the dated timeline; the full narrative with sources is preserved in Appendix C.

3.4.2 Table 2. Milestones in the standardization of physics, 1791–2019

Year	Event	Layer(s) moved	Source
1791–1799	Metric system reported to French National Assembly (1791); Delambre–Méchain survey (1792–98); Mètre des Archives + kilogram deposited (22 June / 10 Dec 1799)	L0, L2	Alder, <i>The Measure of All Things</i> (2002); Wikipedia “History of the metre”; NIST timeline
1832	Gauss: absolute measurement of magnetic field in mm–g–s	L0, L1	“A Brief History of the SI Units” (Univ. Illinois); arXiv:1506.01951
1861–1874	BAAS Committee on Electrical Standards (Thomson, Maxwell, Joule); CGS system	L0, L2	arXiv:1512.07306; arXiv:2411.09568; Sizes.com
1875	Metre Convention signed 20 May by 17 nations; BIPM/CGPM/CIPM created	L2, L8, L9	BIPM “Metre Convention”; Milton (2016)
1887	PTR founded (first national metrology institute)	L9	PTB, “PTR and PTB: History of an Institution”
1889	1st CGPM sanctions international prototype metre & kilogram	L0, L2	BIPM; “A Brief History of the SI Units”
1893	<i>Physical Review</i> founded	L4	APS journals history
1900/1901	NPL (1900), NBS (1901; → NIST 1988) founded	L9	NPL; ACS/NIST history
1901	Giorgi proposes MKS + electrical base unit (MKSX)	L0, L2	IEC “History of the SI”; Britannica
1921	Metre Convention amended to cover all physical (incl. electrical) units	L2, L8	BIPM; gum.gov.pl
1922	IUPAP founded (Brussels, 13 members)	L3, L8	IUPAP history
1929	Birge, “Probable Values of the General Physical Constants,” <i>Rev. Mod. Phys.</i> 1, 1 (first article, first issue)	L3	APS journals; Physics Today RMP history
1932	Birge ratio for consistency of least-squares data	L6	Hedges (1987); PDG usage
1948–1954	9th CGPM requests practical system (1948); 10th CGPM adopts six base units (1954)	L0, L2	BIPM; sizes.com
1955	OIML created by convention	L8, L9	OIML
1957–1968	Gell-Mann–Rosenfeld review + Barkas–Rosenfeld UCRL-8030 (1957); PDG branded 1968	L2, L3, L6	CERN Courier (2017); symmetry magazine; CERN SIS
1960	11th CGPM names the SI	L0, L2	BIPM; Britannica; IEC
1966–1969	CODATA founded (1966); Task Group on Fundamental Constants established (1969)	L3, L8	NIST CUU; CODATA
1973	First CODATA recommended constants set	L3	<i>Rev. Mod. Phys.</i> 59, 1121 (1987); CODATA
1974	SPIRES-HEP database at SLAC	L4	Cambridge European Review; SLAC
1975	Rosenfeld, “The Particle Data Group: Growth and Operations,” <i>Annu. Rev. Nucl. Sci.</i> 25, 555	L6	DOI 10.1146/annurev.ns.25.120175.003011
1978	ISO Guide 25 (precursor of ISO/IEC 17025)	L9	LIMSWiki; ResearchGate “ISO/IEC 17025: History”
1980	CIPM Recommendation INC-1 (foundation of the GUM)	L1	JCGM 100:2008; Springer (VIM history)
1984	VIM first edition	L3	Springer, “The 2012 VIM”
1986/1998	CODATA 1986; CODATA 1998 initiates 4-year cycle	L3	<i>Rev. Mod. Phys.</i> 59, 1121; arXiv:1203.5425
1991	arXiv conceived (Ginsparg); SPIRES first NA web server	L4	arXiv physics/0007040; Cambridge European Review
1993	GUM first published (ISO/IEC Guide 98)	L1, L5	ISO 45315; BIPM JCGM 100:2008
1999	ISO/IEC 17025 1st edition; CIPM MRA signed 14 Oct (38 NMIs); MCO loss (unit-interface failure)	L9	ISO; BIPM CIPM-MRA; NASA MIB Phase I Report
2009/2016	ISO 80000-1 (2009); PhySH launched (2016)	L3	Wikipedia “ISO/IEC 80000”; Wikipedia “PhySH”
2010	INSPIRE-HEP launched (succeeds SPIRES)	L4	Wikipedia “INSPIRE-HEP”; INSPIRE
2018/2019	26th CGPM redefines SI on fixed constants (voted 16 Nov 2018;	L0	BIPM Resolution 1 (26th CGPM); NPL

3.4.3 When did physics cross the layer threshold?

Dating each layer's crossing of ~60/100 gives:

- **L0 Measurand:** crossed ~1889–1960 (international prototypes → SI named); modern high (constant-based) 2019.
- **L1 Instrument/uncertainty grammar:** crossed 1993 (GUM); modern high with JCGM suite (2008–).
- **L2 Identity:** crossed ~1889–1960 for units; PDG identity from 1957/1968.
- **L3 Vocabulary & reference data:** crossed 1969 (CODATA Task Group) for constants; 1984/2009 for formal vocabularies.
- **L4 Study record/provenance:** crossed ~1974–1991 (SPIRES, arXiv).
- **L5 Reporting:** crossed 1993 with GUM reporting rules; reinforced by journal norms.
- **L6 Synthesis/grading:** crossed with the PDG scale factor convention (late 1950s–1970s) and the 5 σ convention (formalized culturally by the 1990s).
- **L7 Translation/scope:** matured with the SI Brochure editions and ISO 80000 (2009).
- **L8 Governance/incentives:** crossed 1875 (treaty), extended 1921, CODATA/OIML mid-century.
- **L9 Traceability:** crossed 1999 (ISO/IEC 17025 + CIPM MRA), building on NMIs from 1887.

Applying the paper's weakest-layer rule, a field satisfies the first condition of the functional floor only when its last layer meets the 60-point layer threshold. The candidate answers are ~1875 (units treaty), ~1929–1969 (reference-data governance), ~1960 (SI named), and ~1993–1999 (uncertainty grammar + traceability chain). The defensible single answer is **the 1990s** — concretely 1993–1999 — when GUM (1993), ISO/IEC 17025 (1999), and the CIPM MRA (1999) closed the uncertainty and traceability layers that had lagged the units treaty by roughly 60–120 years. Physics was not effectively standardized in 1875. Units led; reference-data governance (Birge 1929 → CODATA 1969) and the uncertainty/traceability grammar (INC-1 1980 → GUM 1993 → MRA 1999) followed one to two human generations later. These layer-crossing dates are interpretive judgments about continuous processes, not discrete events (see Limitations).

3.4.4 Implications for nutrition's timeline expectations

1. **Standardization is measurand-first and bottom-up.** Physics fixed what it was measuring (units, prototypes) long before it standardized how well (uncertainty) or how comparably (traceability). Nutrition should expect its measurand and method layers (L0–L1) to move first, and its synthesis/traceability layers to lag by design, not by failure.
2. **Whole-stack standardization is a decadal-to-century process.** It took ~125 years from the Metre Convention (1875) to a complete traceability chain (1999), and ~90 years from the first constants adjustment (Birge, 1929) to a constant-based SI (2019). Expectations for a fully standardized nutrition evidence stack should be measured in decades, not years.
3. **Each layer was driven by a distinct, felt practical crisis.** Telegraphy forced electrical units; WWII/atomic-era precision forced constants governance; globalized trade forced the MRA. Nutrition layers will likely move only where an analogous concrete crisis (e.g., irreproducibility scandals, regulatory liability) creates demand.
4. **Governance stayed consensus/treaty at the units tier and never touched research conduct.** The CGPM/CIPM system standardizes references and traceability, not how physicists design or run experiments. Nutrition should not expect a single governance body to standardize research conduct; the physics analogue standardized the shared references, leaving conduct to journals and communities.
5. **Single-layer adoption on a 5–15 year timescale is historically reasonable; whole-stack is not.** A reporting standard such as an “MI-Nutrition” checklist can plausibly propagate through journals in 5–15 years, matching MIAME's uptake (published December 2001; adopted by major journals by summer 2002 and broadly required by ~2006). But that is one layer (L5). Expecting the entire nine-layer stack to reach the layer threshold at every layer on that timescale contradicts the physics record.
6. **The weakest layer sets the field's status.** Because the layer threshold is applied under a weakest-layer rule, nutrition's effective standardization date will be governed by its slowest layer — most likely traceability/synthesis — not by early wins in units or vocabulary. Progress reports should track the lagging layer explicitly.

4. Diagnosing nutrition

Applying the framework, nutrition's profile is not jagged but uniformly low: no layer reaches 60, the highest is Instrument/Method at 55, and Scope falls to 20. Because standardization compounds downward, the collapsed layers cap the field — and here there is no high layer left for them to cap.

The Layer-0 ceiling

Nutrition's foundational problem is that many of its core “measurands” are not measurands. “A tomato” is a category spanning cultivars, ripeness, soil, and season; “dietary fiber” is defined by the assay used to measure it, not independently of it. This is not a rhetorical flourish — it is the metrological point that a quantity whose value depends on the measurement procedure is not a well-defined measurand in the VIM sense. Two consequences follow.

The first is what we call the “**well-labeled garbage**” problem: no amount of downstream rigor repairs an ill-defined measurand. You can assign a tomato a persistent identifier (L2), analyze it with a validated method (L1), and grade the resulting evidence with GRADE (L6), and still have a precisely recorded, well-graded number for a quantity that does not denote a stable thing.

One clarification prevents an over-reading of this ceiling. Category-level measurands are not disqualifying, and nutrition is not uniquely afflicted by them; they are workable in exactly the way physics works with noisy quantities — by specifying conditions, declaring uncertainty, and calibrating against reference measurements. Nutrition already possesses each of these tools: USDA Foundation Foods publishes sample counts, standard deviations, ranges, and acquisition years for analytical values; recovery biomarkers (doubly labeled water, urinary nitrogen) calibrate self-reported intake; and the NCI measurement-error methods propagate the resulting error into effect estimates. The Layer-0 ceiling therefore bounds achievable precision, not scientific legitimacy. The failure this paper diagnoses is not that nutrition uses categories, but that the spread behind each category is routinely discarded rather than carried.

The second is **the Atwater illustration**. Metabolizable energy of foods is conventionally computed with the Atwater factors, over a century old. Controlled-feeding studies at the USDA Beltsville Human Nutrition Research Center found the Atwater system overestimates the metabolizable energy of almonds by 32% and of walnuts by 21%, while pistachios are overestimated by only about 5%. As Novotny, Gebauer and Baer (2012, *AJCN*) reported for almonds: the measured energy content was “4.6 ± 0.8 kcal/g, which is equivalent to 129 kcal/28-g serving” — significantly less than the 6.0–6.1 kcal/g conventionally assumed, a 32% overestimation. The label number and the measured number diverge because the “energy content of a food” is not a fixed property of the food alone; it depends on food structure, processing, and the eater's digestion. Even the widely cited claim that garden-crop nutrient levels have declined since 1950 rests on this ceiling: Davis, Epp and Riordan (2004) reported statistically significant median declines in six nutrients across 43 garden crops between 1950 and 1999, ranging from 6% for protein to 38% for riboflavin, but the authors themselves cautioned that changes in cultivars and in analytical methods over 50 years cannot be cleanly separated from real change — an L0/L1 confound, not a settled fact.

Unresolved entity identity (L2)

Packaged identity is solved; generic identity is not. GSI's GTIN gives packaged foods a genuinely global, stable identifier, and USDA FoodData Central, LanguaL and EFSA's FoodEx2 are mature, maintained description systems. The score is low anyway because the hard case is unsolved: there is no universal stable identifier for a generic unbranded food — no GTIN for “a raw apple” — and cross-references between registries (FDC ID ↔ LanguaL ↔ FoodEx2 ↔ FoodOn) are partial and hand-curated. The Periodic Table of Food Initiative published a

curated list of 1,650 foods precisely because no canonical list existed. Mature systems that do not resolve the field's basic unit do not make an identity layer. This is the clearest case in the stack of the construct stated in §2.2: the existence of standards is not standardization.

Fragmented reference data (L3)

Reference data in nutrition are plural and unharmonized. An integrative review of the state of food-composition databases (Brinkley et al., 2025, *Frontiers in Nutrition*) evaluated 35 data attributes across 101 databases from 110 countries and found that only **17 of 101 (17%)** satisfy all 13 FAIR criteria evaluated. Findability is universal — every database carries a persistent identifier — and is also the cheapest criterion, asking only for a URL or a DOI. The aggregated scores behind it are accessibility 30%, reusability 43% and interoperability 69%. Quality tracks country income: databases from high-income countries showed greater inclusion of primary data, web interfaces, more regular updates and stronger FAIR adherence, and every database meeting the full FAIR criteria is maintained by a high or upper-middle-income economy. The consequence is measurable: choosing a different food-composition database changes calculated nutrient intakes for the same dietary data (documented, e.g., in the EPIC cohort by Van Puyvelde et al., 2020, *Nutrients*, and in a Swedish–Finnish comparison by Hakala et al., 2003, *Eur J Clin Nutr*). This variability is precisely why a 2025 call to action in the *American Journal of Clinical Nutrition* (Blumberg et al.) argued for community-driven minimum-information standards for food composition data, noting that historical standards have provided guidelines rather than programmatically enforced schemas for data reporting.

Vocabularies are the partial bright spot, though they too sit in the fragmentary band. FoodOn (Dooley et al. 2018, *npj Science of Food*) is a farm-to-fork food ontology; LanguaL, EFSA's FoodEx2, and the FAO/INFOODS tagnames provide additional structured vocabularies; CDNO covers nutritional components. But plurality without a pivot is itself the L3 problem: multiple good vocabularies that do not fully interoperate. A formal standard for the reference layer does exist and the score reflects it — EN 16104:2012 (CEN/TC 387) specifies semantics, structure, controlled vocabularies and an XML interchange for food data. It does not lift the layer higher because it is a paid CEN document with compiler-culture rather than universal uptake: existence without reach, which is what this scale measures.

The measurement layer (L1) shows the same pattern of “standards, plural.” Dietary fiber can be measured by several coexisting AOAC Official Methods (985.29, 991.43, 2001.03, 2009.01, 2011.25, 2017.16, 2022.01, among others), because the Codex Alimentarius definition of fiber admits multiple validated methods per analyte, and different methods yield systematically different fiber values for the same food (McCleary, 2023, *J AOAC Int*).

The strongest counter-argument: the Nutrition Care Process

The strongest objection to this diagnosis comes from clinical dietetics, and it should be stated before it is answered. Nutrition does have a standardized care process with a controlled language: Hakel-Smith and Lewis (2004) describe a standardized nutrition care process and language developed so that dietetic practice could be documented in comparable terms across practitioners and settings. It is a real standard, it was adopted, and it does the thing this paper says nutrition does not do — it fixes a vocabulary and a record structure, and gets them used.

It does not lift the layer scores, and the reason is instructive. The Nutrition Care Process standardizes *the clinical encounter*: the assessment, the diagnosis, the intervention, the monitoring. What it does not standardize is the identity or the composition of the food. A dietitian's record can be perfectly conformant to it and still name a food that has no stable identifier (L2), draw its nutrient values from an unversioned database (L3), and carry no imputation flag forward (L9). The standard sits above the layers diagnosed here and inherits their weaknesses rather than repairing them. That is the compounding principle of §2 applied to a real case, and it is the clearest available demonstration that a well-adopted standard at one level does not rescue an unstandardized level beneath it.

Thin reporting, synthesis, and study-record layers (L4, L5, L6)

At L4, the gap is the study record rather than provenance as such. No machine-readable study-record standard carries provenance: a 2025 survey of provenance, assertion, and evidence ontologies (Chhetri et al.) catalogued numerous general provenance ontologies but found none purpose-built for nutrition. The food-value layer is better served — EN 16104/EuroFIR documents how a number was obtained — but nothing carries that documentation into a record of the study that used it. The cost surfaces as manual re-harmonisation: consortia routinely flatten and cross-walk cohorts by hand to make them comparable.

At L5, a nutrition-specific reporting checklist exists — STROBE-nut (Lachat et al. 2016, *PLoS Medicine*), a 24-item extension of STROBE for nutritional epidemiology — but its uptake has been limited and depends on journal enforcement. Nutrition journals endorse it; none gates publication on it the way *Nature* gated MIAME from December 2002.

At L6, the field's grading gap is stark. In a methodological survey of 800 systematic reviews in the ten highest-impact nutrition journals (2015–2019), Werner et al. (2021, *J Clin Epidemiol*) found that “Forty-seven SRs (5.9%) rated the outcome specific certainty of evidence”; among the 465 certainty-of-evidence outcome ratings they identified, the ratings ranged from very-low (28.8%) to low (41%), moderate (26.5%), and high (3.7%). Grading is both rarely applied and, when applied, rarely reaches high certainty.

The empty translation layer (L7)

Nutrition's deepest collapse is at Scope. Effect estimates are routinely stated without the population, exposure range, and outcome definition to which they apply being bound into the estimate itself. A result obtained in one population and exposure range is silently carried into guidance for another. The clearest recent illustration is ultra-processed food. The 2025 Dietary Guidelines Advisory Committee conducted a systematic review on dietary patterns with varying amounts of ultra-processed foods and growth, body composition, and obesity risk, and graded the evidence “Grade Not Assignable” — unable to draw a conclusion, citing among other things critical limitations and inconsistency in the body of evidence. The exposure (UPF) is defined differently across studies — the NOVA system is the most common but not the only one — and the FDA and USDA jointly opened a Federal Register docket (FDA-2025-N-1793, published 25 July 2025 at 90 FR 35305) requesting information to develop a uniform definition, explicitly because no single universally accepted definition exists. The scope of the claim (“processed how, defined by whom, in which population”) is not bound to the claim, so the evidence cannot be synthesized. That is an L7 failure, and it caps L6.

An independent line of argument reaches an adjacent conclusion from inside the field. López-Moreno, Quesada-Fernández and López-Gil (2026), writing in *Advances in Nutrition*, categorise eight recurrent inferential fallacies in nutrition research — among them an *exposure contrast fallacy* and a *replacement fallacy* — and argue that analytical outputs may retain statistical coherence while failing to correspond to clearly defined causal contrasts or meaningful real-world dietary interventions. Their remedy is greater alignment between estimands, exposure definitions, comparators, and causal interpretation.

The same gap shows where claims run ahead of the evidence that would bound them: Bailey and Stover (2023) argue that precision nutrition's promises currently exceed what the science supports. In the vocabulary used here that is a scope failure — a claim carried beyond the population and exposure range in which it was established, with nothing in the record to stop it.

That diagnosis and this one are at different levels and they compose. They describe how inference distorts *even when accepted methods are correctly applied*; the nine-layer framework asks why the field has no infrastructure in which the contrast, the comparator and the population could be *specified* in the first place. A misalignment that must be caught by the reader's judgment is a Layer 7 failure by construction: there is no field in the study record where the answer could have been written down. Their fallacy catalogue is, on this reading, the downstream symptom of the Layer 0, Layer 2 and Layer 7 gaps diagnosed here — which is also why it is a catalogue of reasoning errors rather than a specification. The convergence is noted as convergence; the framework vocabulary used in this paper is its own and is not drawn from that work.

Traceability: capture is not the failure (L9)

No chain runs from a food measurement through the study to the guideline. The failure is **carry-forward, not capture** — and that holds worldwide, not in one region only. Imputation status (measured versus borrowed, calculated or imputed) is recorded by the major compilers on both sides of the Atlantic and by the global guidance. EuroFIR's Acquisition Type, Value Type and Method Type descriptors document it, with a coded case for a value borrowed from another food (Westenbrink et al., 2021, *J Food Compos Anal*), harmonised across 26 European databases (Westenbrink et al., 2019). USDA FoodData Central ships a dedicated `food_nutrient_derivation` table — its own field descriptions define it as the procedure

indicating how a food nutrient value was obtained, with a derivation code where A means analytical and a source code where 4 means calculated or imputed — carried at both the food-nutrient and the FNDDS ingredient level. FAO/INFOODS documents borrowed values globally.

That provenance is captured at the compiler and then dropped: database version and imputation flag are almost never carried into the study, the synthesis or the guideline, so a downstream number cannot be traced to its origin even though its origin *was written down*. What has no standard is the chain end to end, not the capture at the top of it. The distinction matters for remedy: the intervention required is not a new compiler descriptor — those exist and are good — but a record that carries the descriptor forward.

Summary: how the failures compound

The failures of this section are not independent defects; the weakest-layer rule makes their joint effect simple to state. The Layer-0 ceiling caps how good nutrition's numbers can ever be; the Layer-2, Layer-3 and Layer-4 collapses discard the identity and the lineage — which food, which database version, which imputation status, which structured study record — that would let anyone determine how good a given number actually is; and the Layer-7 collapse discards the scope that would say where the number applies. The last of these is independently fatal: even a stable measurand and perfect records would still yield guidelines untethered from their evidence if the population and contrast are not bound to the claim. A stack with one empty layer is an empty stack for any downstream consumer who needs to traverse it end to end. That is why the remedy in Section 5 is stated as a floor to be crossed at every format layer, not an average to be raised.

5. Path forward

The functional floor

Because standardization compounds downward, the right target is not to maximize the average score but to raise the floor. We propose a **functional floor** definition (a proposal of this paper, not a sourced threshold): a field reaches a functional floor when no format layer scores below 60/100 and every reported effect estimate carries a declared exposure-error term. The 60 threshold is a deliberate, defensible proposal, not an empirically derived constant (see Limitations); its role is to force attention onto the weakest layer rather than to certify a particular number.

Two things follow, and they must not be conflated. The first condition is a per-layer test: we call 60 the **layer threshold**, and a field can be scored against it layer by layer and the results counted. The second condition is a field-level practice and is not scored in Table 1 at all. The **functional floor** is the conjunction of the two, and it is reached or not reached *by a field* — never by a layer. Earlier versions of this paper described fields as clearing the floor “at” a number of layers, which treats a conjunctive field-level predicate as a per-layer count; that wording is corrected here.

On that reading: **nutrition meets the layer threshold at no layer**, its highest score being 55, so it fails the first condition and does not reach the functional floor. **Physics meets the layer threshold at all ten layers**, and it satisfies the analogue of the second condition: the GUM (1993) made a stated uncertainty part of every reported result, and ISO/IEC 17025 with the CIPM MRA (1999) closed the traceability chain that makes such statements comparable across laboratories (§3.1, §3.4). On that basis physics reaches the functional floor.

The two are analogues, not the same object, and the distinction should be kept. A GUM measurement uncertainty quantifies how well an instrument determined a quantity on a defined measurand. An exposure-error term quantifies how badly a self-reported or otherwise indirect exposure measure tracks true intake in a population. They play the same structural role — both bind the reliability of a number to the number — but they are not interchangeable, and physics having the first does not mean nutrition's task is to import it. The nutrition analogue already exists in its own tradition: the recovery biomarkers and measurement-error methods described below. The layer counts are computed from Table 1 rather than asserted; the second condition is established narratively and dated in §3.4, not scored.

The second half — a declared exposure-error term on every estimate — directly imports the physics lesson. Physics binds a regime of validity to every law; the nutrition analogue is to bind an explicit statement of exposure measurement error (and its population and range) to every effect estimate, so that scope travels with the claim. Established nutritional-epidemiology tooling already supports this: the National Cancer Institute's measurement-error methods and recovery biomarkers (doubly labeled water for energy, urinary nitrogen for protein) exist precisely to quantify the exposure error that a declared error term would surface.

The keystone intervention: a machine-readable StudyRecord via MI-Nutrition

We propose one keystone intervention that moves several layers at once. It is a machine-readable **StudyRecord** governed by a minimum-information checklist we call **MI-Nutrition** (a proposal of this paper). Modeled on the minimum-information-standard tradition, MI-Nutrition specifies a required field set for any nutrition study. The full set is published in the schema rather than enumerated here; the five fields below are the ones that carry this paper's argument, and all five are required rather than optional:

1. the food-composition database and version used, with resolvable identifiers;
2. **imputation flags** marking which nutrient values were analytically measured versus borrowed, calculated, or imputed;
3. a **population fence**: an explicit, structured statement of the population to which the estimate applies (age, sex, health status, baseline exposure range);
4. **value-lane typing**: typing each contrast by whether it addresses prevention, treatment, or reversal of a condition, and typing the comparison itself, so that estimates are not silently pooled across incommensurable questions;
5. a **study value hypothesis**: one falsifiable sentence naming the decision the study was designed to inform, as distinct from its scientific hypothesis, so that a result is not later read as evidence for a question it was never designed to answer. It names the decision, not the population — the population is field 3's job, and the two are required to stay distinct, because the claim that each is a separate residual depends on it.

What is new here, and what is not. These five fields do not carry equal novelty, and v1.4 narrows the claim. Tested against ONS, STROBE-nut, EN 16104/EuroFIR, FAO/INFOODS, CDISC SDTM and HL7 FHIR, the imputation field is an *exact match* to EN 16104/EuroFIR's Acquisition/Value/Method Type descriptors — this is convergence on an existing standard, not an invention, and it should be adopted rather than reinvented. The population fence is a *close match* to STROBE Item 21, and its contribution is to make structured and declarative what that guidance leaves as free text. Only **value-lane typing** and the **study value hypothesis** have no counterpart in any of the six standards examined. The residual is smaller than v1.3 implied and stronger for it: both genuinely novel fields map to L7 Translation/Scope, nutrition's lowest-scoring layer at 20. The one place this framework invents something is the one place no upstream standard reaches. The *terms* used here — population fence, value lane, typed contrast, study value hypothesis — are coinages of this paper; coinage of a term does not imply novelty of the underlying field, and for the population fence it does not.

The design point is which layers this moves. The StudyRecord is the Layer 4 remedy: a machine-readable study record carrying provenance is precisely what §4 finds the field lacks, and the checklist is the thing that would make one exist. It binds Layer 7, because the population fence, the value lanes and the study value hypothesis put scope inside the estimate rather than beside it. And it closes Layer 9's carry-forward gap, because a versioned database identifier and an imputation flag travelling in the record are the missing hop between the compiler and the guideline — the compilers already capture imputation status, and what the StudyRecord adds is a place for it to land downstream.

Those three layers — L4 at 30, L7 at 20, L9 at 30 — are three of the four lowest scores in the nutrition column. The fourth, L0 at 25, is the measurand ceiling, and no record format repairs it; that is the boundary of what this intervention can claim. Layer 3 improves as a consequence rather than as a target: naming the database and its version with resolvable identifiers makes reference data more usable without making it less plural, which is the actual L3 problem and not one a study record can solve.

Why community adoption, not statute

The precedent that matters is **MIAME** (Brazma et al. 2001, *Nature Genetics*), the Minimum Information About a Microarray Experiment. MIAME succeeded not because a law required it but because journals adopted it as a condition of publication and the community repositories (GEO at NCBI, ArrayExpress at EBI) required MIAME-compliant deposit. Within a few years, MIAME-compliant deposit became a near-universal norm across the field. The Genomic Standards Consortium's MixS did the same for sequence metadata. This is the adoption path we recommend for MI-Nutrition: journal and repository adoption, not statutory mandate.

Two cautionary precedents reinforce the choice. First, **statutory mandates without aligned incentives underperform**. FDAAA 801 legally requires results of applicable clinical trials to be posted to ClinicalTrials.gov within a year; DeVito, Bacon and Goldacre (2020, *The Lancet*) found that only 40.9% of due trials reported within the one-year deadline despite the legal requirement. A statute did not produce compliance. Second, the **CDISC/FDA single-gatekeeper model is the wrong precedent for nutrition**, because it relies on a single regulatory gatekeeper (the FDA) and a product-approval incentive (a company wants its drug approved). Nutrition has neither a single gatekeeper nor a product-approval choke point, so a gatekeeper model has no lever to pull. The comparison also corrects a common misconception: pre-CDISC pharmaceutical submissions were not “static PDFs” — SAS Transport (XPORT) files were the established data format — so the CDISC story is one of format upgrading under a gatekeeper, a condition nutrition lacks.

The right analogues for nutrition are therefore **MIAME + the labeling-statute precedent**: the Nutrition Labeling and Education Act of 1990 (NLEA) shows that nutrition already accepts standardized formats where a clear point of enforcement exists (the package label), while MIAME shows that community/journal adoption can standardize research records where no such enforcement point exists. MI-Nutrition belongs to the MIAME lineage.

There is momentum to build on. The 2025 *AJCN* call to action for minimum-information standards for food-composition data, the emergence of proposed reporting and reproducibility checklists in adjacent nutrition-metabolism work, the GLIM criteria for malnutrition diagnosis (in a 2023 global survey of more than 1,500 respondents, 25% reported that GLIM had been implemented and 20% that implementation was in process, per Jensen et al.'s 2025 five-year update in *JPEN*; and Japan's Diagnosis Procedure Combination reimbursement system began encouraging routine GLIM-based assessments in June 2024), and USDA's FoodData Central Foundation Foods (which publish sample size, standard deviation, min/max, and year for analytical values) all point the same direction: toward versioned, provenance-carrying, scope-bound records.

6. Conclusion

Nutrition science is not failing because food is variable; food is supposed to be variable, and quantifying that variation is the job. Nor is it wholesale unstandardized — mature artifacts exist, most visibly the food ontologies, reference databases, and validated AOAC methods behind them. What it lacks is any layer where those artifacts have consolidated into one standard the field actually uses: no layer meets the 60-point threshold. It is failing at specific, diagnosable format layers: an ill-defined measurand ceiling (L0), unresolved entity identity (L2), fragmented reference data (L3), an absent structured study record (L4), and above all an empty translation layer (L7) where the scope of a claim is not bound to the claim.

The physics control case shows that near-total standardization is achievable without a reporting checklist and without a research statute, because scope discipline can be linguistic and cultural rather than procedural. The keystone move for nutrition is a machine-readable StudyRecord, governed by an MI-Nutrition minimum-information checklist that makes database version, imputation flags, population fence, value-lane typing, and the study value hypothesis required fields — raising the study-record, translation, and traceability layers together, along the community-adoption path that made MIAME a success.

A machine-readable route (work in progress). The specifications proposed here are served as callable tools over the Model Context Protocol, beside a catalogue of existing standards, so that a language model or a code agent can fetch a specification, resolve an ontology identifier, and validate a study record without a human transcribing anything. Validation returns a computed result from each specification's own reference validator, imported rather than reimplemented; a field declared absent is a conforming declaration, not an error. This is beta, and **it is not adoption**. All four specifications are v0.1 Draft/RFC, every response says so, and the adoption register (Appendix F) records this project as its own first adopter at *piloting* rather than *implemented* for that reason. It is offered as a route rather than a result: the cost of conforming to a standard falls when the standard is callable, and the layers where nutrition scores lowest are the ones whose fields a machine could fill.

The invitation of this working paper is correspondingly concrete. Journals and repositories can adopt the StudyRecord fields as a submission option; database compilers can version their releases, flag imputed values, and — the specific ask of §4 — ensure those flags survive ingestion downstream; and researchers can run any arm of the validation protocol in Appendix E: re-score Table 1 against the §2 rubric and the §2.2 construct, apply the stack to an adjacent field, or pilot the MI-Nutrition fields against a sample of published studies, and report where the framework breaks. That is how MIAME's floor was crossed: not by mandate, but by a community that adopted, tested, and revised the artifact.

Revision history

- **v1.0** (23 Aug 2026): initial working paper with claims register.
- **v1.1** (23 Aug 2026): dated physics history condensed to §3.4 with full narrative preserved in Appendix C; compounding summary added at end of §4.
- **v1.2** (23 Aug 2026): scoring attribution corrected to single-author; Layer-0 ceiling language tightened (precision, not legitimacy); external-validation protocol added to Limitations. Revisions respond to external review.
- **v1.3** (23 Aug 2026): BIPM membership corrected to 68/35 against the live BIPM page; abstract control-case sentence split; Layer-7 “regime of validity” claim given the same epistemic caveat as claim 13 (new claim 31); forward-looking community invitation added to the Conclusion.
- **v1.4** (31 Aug 2026): **Table 1 Layer 2 Identity (nutrition) corrected 70 → 33**, with erratum. Construct definition added as §2.2 and the layer-threshold result stated in §5 — no nutrition layer meets 60, against ten of ten for physics; the “peak at Identity” characterization is removed from the abstract, §4, §5 and §6 accordingly. Citation corrected from the first deposit's version DOI to the concept DOI throughout. Layer 9 rationale corrected: imputation status *is* recorded by EuroFIR, USDA FDC and FAO/INFOODS, and the failure is carry-forward, not capture. Layer 3 corrected against the full text of Brinkley et al. (2025): 17 of 101 databases satisfy all 13 FAIR criteria, findability is universal, interoperability is 69%. Layer 4 rescoped to the study record. EN 16104:2012 named at L3 and L4. MI-Nutrition novelty narrowed from four fields to two. López-Moreno et al. (2026) added at L7 as convergent external evidence. Functional floor separated into a per-layer *layer threshold* and the conjunctive field-level *functional floor*, correcting wording that counted a field-level predicate per layer, with a note that a GUM measurement uncertainty and an epidemiologic exposure-error term are structural analogues rather than one object. MI-Nutrition field list corrected: the study value hypothesis is now specified rather than only named, and the unpublished field count is dropped. Keystone claim corrected: the StudyRecord is credited with Layer 4, which it instantiates, in place of Layer 3, which at 45 is not among the lowest layers; the abstract and conclusion follow. Appendix F's lane counts reconciled to its stated total. Hakel-Smith & Lewis (2004) added to §4 as the paper's strongest counter-argument; Bailey & Stover (2023) added at L7 and Penders et al. (2017) at §1. Score-stability and construct-conformance limitations added; score register, validation protocol and adoption register added as Appendices D, E and F. No score other than Layer 2 changed.

Limitations

- **The comparison scores are ordinal expert judgments, not measurements.** The 0–100 values in Table 1 encode relative standing against a verbal rubric; they were assigned by the author against the cited infrastructure for each layer. They should not be treated as having interval or ratio properties, and small differences between scores are not meaningful.
- **The framework has not been externally validated, and the path to validation is specifiable.** The Nine-Layer Stack is a proposed analytic tool. It has not been tested against independent raters, applied to additional fields, or evaluated for inter-rater reliability. The validation protocol this implies is concrete: (i) independent domain experts re-score Table 1 using the §2 rubric and the §2.2 construct, with agreement summarized by a weighted kappa or intraclass correlation; (ii) the taxonomy is applied to at least two adjacent fields (for example, dietary assessment methodology and clinical nutrition, with metabolomics as a bridge case) to test whether the layers discriminate; and (iii) the proposed MI-Nutrition fields are piloted by extracting them from a sample of published studies to measure field-level presence and extraction reliability. Until (i)–(iii) are done, Table 1 should be read for its relative ordering, not its numbers. See Appendix E.
- **Score stability is now audited.** Between the 15 August 2026 recalibration and the 23 August Table 1, five layers moved. Four are accounted for by the taxonomy migration — eight of the ten layer labels changed — and two of those four crossed a rubric band without that being declared at the time. The fifth, Layer 2, had no explanation and is corrected in this version. Every such move is now recorded, with its reason and its evidence, in a machine-readable `rescore_log` published beside the scores, and a test fails the build if a score moves

- ≥10 points on an unchanged construct without an entry. The framework has still not been externally validated; what has changed is that unexplained movement is now detectable rather than discoverable only by a reader who diffs two versions by hand.
- **The construct was applied to one layer, not ten.** The definition added at §2.2 is general, but only Layer 2 was rescored against it. The remaining layers retain scores assigned before the construct was stated, and a systematic re-derivation against §2 is outstanding. Layer 1 is the most likely to move, for the reason given in §4: multiple validated methods coexist per analyte and yield systematically different values for the same food, which is the same plurality-without-a-pivot defect that lowered Layer 2. Layer 1 is also the current maximum of the nutrition column, so a downward move there would lower the column’s ceiling without changing the floor result.
 - **The functional-floor threshold of 60 is a proposal, not a sourced value.** No empirical work establishes that 60/100 is a meaningful cut-point; the number is a deliberate design choice to focus attention on the weakest layer. The finding that nutrition meets it at no layer is therefore a statement about the column relative to a chosen line, not a measured property of the field.
 - **Novel coinages are proposals of this paper.** “Functional floor,” “population fence,” “value lane,” “value-lane typing,” “MI-Nutrition,” “typed contrast,” “study value hypothesis,” and “StudyRecord” (in this specific sense) are introduced here as proposals; none is attributable to an external source, and none has been adopted by any standards body. Coinage of a term does not imply novelty of the underlying field — see §5, where two of the five MI-Nutrition fields are shown to have counterparts in existing standards.
 - **Education and public transmission are deliberately out of scope.** How nutrition evidence reaches consumers, clinicians, and the press is a real problem but is downstream of the evidence-production layers analyzed here.
 - **Physics scores are the author’s estimates.** Even where physics infrastructure is documented (SI, GUM, PDG, arXiv), the score assigned to each layer remains a judgment.
 - **Layer-crossing dates are interpretive.** Layer-crossing dates are interpretive judgments about continuous, overlapping historical processes rather than discrete events, and reasonable historians may place a given layer’s ~60/100 threshold a decade earlier or later than we do.
 - **Membership counts are point-in-time.** The BIPM membership figures were verified against the live BIPM member-states page on 23 August 2026 (68 Member States, 35 Associates); because accessions continue, the counts are dated in the text and will drift.
 - **Two standards are cited for existence and reach, not for their clauses.** EN 16104:2012 is a paid CEN document and was not read in full; LanguaL’s registry returns HTTP 455 to automated requests and was not independently verified. Both are cited for what they are and how far they reach, never for the content of their provisions.

Appendix A: Claims register

Each numbered empirical claim in the paper is mapped to its primary source. Claims labeled (proposal) are coinages of this paper and are not empirical claims. Claims 32–41 were added in v1.4.

1. **2019 SI redefinition fixes all seven base units to seven defining constants with zero uncertainty** — BIPM, SI Brochure, 9th ed. (2019); CGPM 26th meeting Resolution 1 (bipm.org).
2. **Metre Convention signed 20 May 1875; 68 Member States and 35 Associate States and Economies** (BIPM member-states page, verified 23 August 2026; Albania acceded 4 June 2026, Peru 11 August 2026) — bipm.org/en/member-states.
3. **GUM (JCGM 100:2008) and VIM (JCGM 200:2012) definitions of measurand and metrological traceability** — BIPM/JCGM (jcgmbipm.org).
4. **ISO/IEC 17025:2017 laboratory accreditation; ISO 80000 quantities and units** — ISO (iso.org).
5. **CODATA recommended constants via four-year least-squares adjustment** (2014/2018/2022), published by NIST — CODATA (codata.org); NIST (physics.nist.gov/cuu).
6. **PDG Review of Particle Physics averaging and scale-factor procedure** — Particle Data Group (pdg.lbl.gov).
7. **PhysSH introduced 2016 replacing PACS** (final version PACS 2010) — American Physical Society (physsh.org).
8. **arXiv founded 1991** (Ginsparg; now Cornell); INSPIRE-HEP (CERN/DESY/Fermilab/SLAC); HEPData (Durham/IPPP) — arxiv.org; inspirehep.net; hepdata.net.
9. **FITS first defined 1981, IAU-endorsed 1982** — NASA GSFC FITS Standard (fits.gsfc.nasa.gov); IAU FITS Working Group.
10. **CIPM MRA (1999)** — BIPM.
11. **5-sigma discovery threshold** ($p \approx 3 \times 10^{-7}$), applied to 2012 Higgs discovery — CERN/ATLAS.
12. **OIML legal metrology as the commerce-tier statute** — OIML (oiml.org).
13. **Physics has no CONSORT-analogue mandatory reporting checklist** — no such standard identified; argument from absence.
14. **101 food-composition databases across 110 countries, 35 data attributes; 17 of 101 (17%) satisfy all 13 FAIR criteria; findability 100%, interoperability 69%, reusability 43%, accessibility 30% (aggregated); FAIR adherence tracks country income** — Brinkley et al. (2025), *Frontiers in Nutrition* 12:1552367 (DOI 10.3389/fnut.2025.1552367). *The 17/101 count is stated in the paper’s Results prose, not in its abstract, and the 17 databases are identified in its Table 1. Figures corrected in v1.4 against the full text, read 31 August 2026; v1.3 characterized interoperability as weak, which the paper’s own aggregated scores do not support.*
15. **Nutrient-intake estimates vary with database choice** — Van Puyvelde et al. (2020), *Nutrients* 12:2906 (EPIC cohort); Hakala et al. (2003), *Eur J Clin Nutr* 57:1035–1044.
16. **Atwater overestimates metabolizable energy: almonds 32% (129 vs 168–170 kcal/28-g serving), walnuts 21%, pistachios ~5%** — Novotny et al. (2012), *AJCN* 96(2):296–301; Baer et al. (2016), *J Nutr* 146:9–13; USDA-ARS Beltsville.
17. **Davis et al. (2004): median declines from 6% (protein) to 38% (riboflavin) across 43 garden crops 1950→1999, with the authors’ own cultivar/method caveat** — *J Am Coll Nutr* 23(6):669–682.
18. **Werner et al. (2021): 47 of 800 nutrition SRs (5.9%) rated outcome-specific certainty with GRADE; of 465 ratings, very-low 28.8%, low 41%, moderate 26.5%, high 3.7%** — *J Clin Epidemiol* (PMID 33588024).
19. **STROBE-nut: 24-item extension, limited uptake** — Lachat et al. (2016), *PLoS Medicine* 13(6):e1002036.
20. **Multiple coexisting AOAC dietary fiber methods; Codex admits multiple methods per analyte** — McCleary (2023), *J AOAC Int* 106(4):917.
21. **FoodOn ontology** — Dooley et al. (2018), *npj Science of Food* 2:23 (DOI 10.1038/s41538-018-0032-6).
22. **Chhetri et al. (2025): provenance/assertion/evidence ontology survey, none nutrition-specific** — PMC12376154.
23. **2025 AJCN call to action for minimum-information standards for food-composition data** — Blumberg et al. (2025), *AJCN* (DOI 10.1016/j.ajcnut.2025.06.027).
24. **2025 DGAC graded UPF–obesity evidence “Grade Not Assignable”** — USDA NESR systematic review (2024); NBK611035.
25. **FDA/USDA UPF definition docket FDA-2025-N-1793**, published 25 July 2025 — Federal Register 90 FR 35305.
26. **MIAME and journal/GEO–ArrayExpress deposit enforcement** — Brazma et al. (2001), *Nature Genetics* 29:365–371; Brazma (2009), TSW.
27. **FDAAA 801 compliance 40.9% on-time despite legal mandate** — DeVito, Bacon & Goldacre (2020), *The Lancet* 395:361–369.
28. **NLEA 1990 as nutrition’s labeling statute** — Nutrition Labeling and Education Act of 1990.
29. **GLIM adoption** (25% implemented, 20% in-process in 2023 survey of >1,500 respondents; Japan DPC reimbursement encouragement from June 2024) — Jensen et al. (2025), GLIM 5-year update, *JPEN* (DOI 10.1002/jpen.2756).
30. **USDA FoodData Central Foundation Foods carry n, SD, min/max, year** — USDA FoodData Central (fdc.nal.usda.gov).
31. **Physics binds “regime of validity” to the statement of its laws (Layer 7)** — disciplinary norm, argument from observed practice; like claim 13, no citable governing standard exists, and the claim’s epistemic status is flagged in §3.2.
32. **GS1 GTIN provides a global stable identifier for packaged foods; PTFI published a curated 1,650-food reference list because no canonical list existed** — GS1; Jarvis et al. (2024), *Nature Food* 5:189–193 (DOI 10.1038/s43016-024-00941-y); EFSA FoodEx2 (DOI 10.2903/sp.efsa.2015.EN-804). *Basis for the corrected Layer 2 score.*
33. **EuroFIR Acquisition/Value/Method Type descriptors record imputation status, including a coded case for a value borrowed from another food** — Westenbrink, Presser, Roe & Ireland (2021), *J Food Compos Anal* 102:103968 (DOI 10.1016/j.jfca.2021.103968).
34. **Harmonised EuroFIR documentation evaluated across 26 European food composition databases** — Westenbrink, Kadvan, Roe & Koroušić Seljak (2019), *J Food Compos Anal* 83:103257 (DOI 10.1016/j.jfca.2019.03.006).
35. **USDA FoodData Central ships a food_nutrient_derivation table defining how a nutrient value was obtained, with derivation code A = analytical and source code 4 = calculated or imputed** — USDA FoodData Central, *Download Field Descriptions* (Oct 2020), read 31 August 2026.
36. **EN 16104:2012 (CEN/TC 387) specifies semantics, structure, controlled vocabularies and an XML interchange for food data** — CEN. *Paid document; cited for existence and reach only, clauses unread (see Limitations).*
37. **LanguaL faceted food-description thesaurus** — languaL.org. *Registry returns HTTP 455 to automated requests; not independently verified (see Limitations).*

38. **López-Moreno, Quesada-Fernández & López-Gil (2026):** eight recurrent inferential fallacies in nutrition research, including exposure-contrast and replacement fallacies; remedy is alignment between estimands, exposure definitions, comparators and causal interpretation — *Advances in Nutrition* (DOI 10.1016/j.advnut.2026.100693). *Cited as convergent external evidence at L7; not a source for this paper’s vocabulary.*
39. **Standardized nutrition care process and language** — Hakel-Smith & Lewis (2004), *JADA* 104:1878–1884 (DOI 10.1016/j.jada.2004.10.015). *Cited in §4 as the strongest counter-argument to this paper’s diagnosis.*
40. **Precision nutrition claims outpace the evidence** — Bailey & Stover (2023), *Annu Rev Nutr* 43:385–407 (DOI 10.1146/annurev-nutr-061021-025153). *Cited in §4 at L7.*
41. **Nutrition science’s difficulties are institutional as well as methodological** — Penders et al. (2017), *Eur J Nutr* 56:2009–2012 (DOI 10.1007/s00394-017-1507-y). *Cited in §1.*

Appendix B: Sources for Section 3.4 and Appendix C (historical milestones)

Alder, K. *The Measure of All Things*. Little, Brown, 2002. (Pre-metric multiplicity: 250,000 different units under some eight hundred names.)

Kula, W. *Measures and Men*. Princeton University Press, 1986 (Eng. transl. of *Miary i ludzie*, 1970).

Delambre–Méchain survey / Mètre des Archives (1799). Wikipedia, “History of the metre”; NIST, “Timeline for the Definition of the Meter,” <https://www.nist.gov/document/museum-timelinepdf>

Gauss, absolute units (1832). “A Brief History of the Development of the SI Units,” Univ. of Illinois; “Babel of Units,” arXiv:1506.01951

BAAS Committee on Electrical Standards / CGS (1861–1874). arXiv:1512.07306; arXiv:2411.09568

Metre Convention (1875). BIPM, “Metre Convention,” <https://www.bipm.org/en/metre-convention> ; M. J. T. Milton, “The Metre Convention and the BIPM” (2016)

PTR / first NMI (1887). PTB, “PTR and PTB: History of an Institution”

NPL (1900), NBS→NIST (1901/1988). NPL, <https://www.npl.co.uk/about-us/national-metrology-institute> ; ACS, “NIST Landmark”; NIST, “From NBS to NIST”

Giorgi (1901), MKS→SI, CGPM history. IEC, “History of the SI,” <https://www.iec.ch/history-si> ; Britannica, “International System of Units”; <https://www.sizes.com/units/SI.htm>

IUPAP (1922). IUPAP, “History,” <https://iupap.org/who-we-are/history/>

Birge (1929). R. T. Birge, “Probable Values of the General Physical Constants,” *Rev. Mod. Phys.* 1, 1 (1929); Physics Today, “The history of Reviews of Modern Physics”

Birge ratio (1932) / scale-factor lineage. Hedges, “How Hard Is Hard Science...”

OIML (1955). Wikipedia, “International Organization of Legal Metrology”

PDG origins (1957/1968). CERN Courier, “The physicist’s guide to the universe” (2017); symmetry magazine, “Particle Data Book”; CERN SIS, “Review of Particle Physics”

CODATA (1966) / Task Group (1969) / adjustments. NIST CUU, “International aspects”; CODATA 2014; Cohen & Taylor, “The 1986 adjustment,” *Rev. Mod. Phys.* 59, 1121 (1987)

SPIRES-HEP (1974) / arXiv (1991) / INSPIRE (2010). Cambridge, “Scholarly Communication in High-Energy Physics”; O’Connell, “Physicists Thriving with Paperless Publishing,” arXiv:physics/0007040; Wikipedia, “INSPIRE-HEP”

Rosenfeld PDG history (1975). A. H. Rosenfeld, *Annu. Rev. Nucl. Sci.* 25, 555 (1975), DOI 10.1146/annurev.ns.25.120175.003011

PDG scale factor (current). Particle Data Group, *Review of Particle Physics*, Introduction/Procedures; Navas et al., *Phys. Rev. D* 110, 030001 (2024).

Franklin, A. *Shifting Standards: Experiments in Particle Physics in the Twentieth Century*. Univ. of Pittsburgh Press, 2013.

5 σ convention. L. Lyons, “Discovering the Significance of 5 sigma,” arXiv:1310.1284 (2013).

ISO Guide 25 (1978) / ISO/IEC 17025 (1999). LIMSWiki, “ISO/IEC 17025”; ISO/IEC 17025:1999.

INC-1 (1980) / GUM (1993) / VIM (1984). ISO/IEC Guide 98:1993; BIPM, JCGM 100:2008; Springer, “The 2012 International Vocabulary of Metrology”

CIPM MRA (14 Oct 1999). BIPM, “CIPM MRA participants”; BIPM, “CIPM MRA: Past, present and future”

ISO 80000-1 (2009) / PhySH (2016). Wikipedia, “ISO/IEC 80000”; Wikipedia, “PhySH”

2019 SI redefinition. BIPM, “Resolution 1 of the 26th CGPM (2018)”; NPL, “The redefinition of the SI units”

Mars Climate Orbiter (1999). Mars Climate Orbiter Mishap Investigation Board, Phase I Report (10 Nov 1999); NASA release 99-134.

MIAME analogy. Brazma et al., *Nat. Genet.* 29, 365 (2001); Brazma, “MIAME – Successes, Failures, Challenges,” PMC5823224.

Note on unverified items: The exact earliest print edition of the PDG *Review of Particle Properties* in which the scale-factor/error-inflation procedure first appears (likely 1969–1973, *Rev. Mod. Phys.*) could not be confirmed from currently accessible online archives and should be checked against a 1970s print edition before final publication. All other dates in this section were verified against the primary or authoritative secondary sources cited above.

Appendix C: Extended historical narrative for Section 3.4

C.1 The measurand and units layer (L0–L2): 1791–1960, completed 2019

The starting condition was extreme heterogeneity. According to Ken Alder’s *The Measure of All Things* (2002), the Ancien Régime of France employed some 250,000 different units of weights and measures operating under roughly eight hundred names, with the same-named unit taking different values by town and by trade; Alder and Witold Kula (*Measures and Men*, 1986) treat this multiplicity as the felt crisis that motivated reform. The metric response was deliberate and dated: the Paris Academy of Sciences reported the decimal system to the National Assembly in 1791; the Delambre–Méchain meridian survey ran 1792–1798; and on 22 June 1799 (with the legal basis fixed 10 December 1799) the platinum Mètre des Archives and the kilogram standard were deposited in the French National Archives.

Standardization of the physical content of units — turning a national artifact into an international, coherent system — took another century and proceeded in identifiable steps: **1832**, Gauss reduced measurement of the Earth’s magnetic field to mechanical units (millimetre–gram–second), the first “absolute” measurement in terms of length, mass, and time. **1861–1874**, driven by the practical crisis of transatlantic telegraphy, the British Association for the Advancement of Science Committee on Electrical Standards (Thomson/Kelvin, Maxwell, Joule) built a coherent system, culminating in the CGS system endorsed around 1873–1874. **1875**, the Metre Convention was signed in Paris on 20 May by representatives of 17 nations, creating the BIPM, CGPM, and CIPM. **1889**, the 1st CGPM sanctioned the international prototype metre and kilogram. **1901**, Giorgi showed that mechanical units plus one electrical base unit could form a coherent four-dimensional system (MKSX). **1921**, the Metre Convention was amended (6th CGPM) to extend beyond length and mass to all physical measurement, including electrical units. **1948–1954**, the 9th CGPM requested a practical system and the 10th adopted six base units. **1960**, the 11th CGPM formally named the *Système International d’Unités*. **2019**, the 26th CGPM (voted 16 November 2018; effective 20 May 2019) redefined the kilogram, ampere, kelvin, and mole in terms of fixed fundamental constants, completing the measurand layer.

The pattern is important: the treaty of 1875 governed the units layer, but each extension (electrical, thermal, luminous) was driven by a distinct practical crisis and arrived decades apart. The measurand layer was not “finished” until 2019.

C.2 The reference-data layer (L3): 1929 → 1969 → 4-year cycle

Governance of the fundamental constants — the shared reference data of physics — lagged the units treaty by more than half a century. The first systematic least-squares review of the physical constants was Raymond T. Birge’s “Probable Values of the General Physical Constants,” which was literally the first article in the first issue of *Reviews of Modern Physics* (*Rev. Mod. Phys.* 1, 1, published 1 July 1929). Birge’s least-squares methodology (refined in his 1932 work) became the standard, but adjustments occurred at irregular intervals through the DuMond–Cohen era of the 1940s–1950s. Institutional governance arrived only when CODATA was founded in 1966 (as an interdisciplinary committee of ICSU) and established its Task Group on Fundamental Constants in 1969. The first CODATA-recommended set appeared in 1973, followed by 1986 and 1998; from 1998 onward the Task Group moved to a regular 4-year cycle. So the reference-data layer began ~1929, acquired durable governance in 1969, and reached its modern cadence only in 1998.

C.3 Identity and vocabulary layers (L2–L3): 1922 → 1957 → 2009/2016

Controlled vocabularies and identity systems developed on separate tracks. IUPAP was founded in 1922 (Brussels, 13 founding member countries) and its SUN(AMCO) commission has coordinated symbols, units, and nomenclature since 1931. Formal quantity/unit vocabularies were codified late: ISO 31 appeared as a family in 1978–1992, and was superseded by ISO

80000, whose Part 1 was published in November 2009. Within particle physics, the identity/vocabulary function is carried by the Particle Data Group, whose lineage runs from the 1957 Gell-Mann–Rosenfeld review and the Barkas–Rosenfeld tables (UCRL-8030, 1957; wallet card 1958); the group branded itself the PDG in 1968. Subject classification for the literature moved from PACS (created by AIP in the 1970s, maintained through PACS 2010) to PhySH (Physics Subject Headings), unveiled by APS in January 2016.

C.4 Study record and provenance (L4): 1893 → 1974 → 1991 → 2010

The provenance layer was built on journal and database infrastructure. *Physical Review* was founded in 1893; abstracting services date from the late nineteenth century. The decisive infrastructure for high-energy physics was SPIRES-HEP, run at SLAC from 1974 as the first comprehensive online catalogue of the HEP literature (and, in 1991, the first web server in North America); Paul Ginsparg’s arXiv preprint server was conceived in 1991; and SPIRES was succeeded by INSPIRE-HEP (launched in beta in 2010). Reaction/measurement data compilation (the Durham HepData effort) grew out of the 1970s reaction-data compilation tradition. The provenance layer thus crossed into robust, queryable form in the 1974–1991 window and matured by 2010.

C.5 Synthesis and grading (L6): Birge ratio → PDG scale factor → 5σ

Physics developed explicit, published conventions for combining discrepant measurements. The statistical root is the Birge ratio (R. T. Birge, 1932) for testing the consistency of least-squares residuals. The PDG operationalized this as its “scale factor”: when N measurements of a quantity are averaged and $\chi^2/(N-1)$ exceeds 1, the input errors are all multiplied by $S = \sqrt{\chi^2/(N-1)}$ so that the rescaled χ^2 equals $N-1$ (PDG, *Review of Particle Physics*, Introduction/Procedures; current edition Navas et al., *Phys. Rev. D* 110, 030001, 2024). This convention grew out of the Rosenfeld-era compilations from the late 1950s onward; Rosenfeld’s own history (1975) and Allan Franklin’s *Shifting Standards* (2013) document its solidification. The parallel discovery convention — the 5σ threshold for claiming a discovery — emerged culturally from the 1960s–1970s bump-hunting era and is critically reviewed by Louis Lyons (arXiv:1310.1284, 2013), who notes that particle physics conventionally requires the background-only fluctuation probability to be no larger than about 3×10^{-7} before a discovery is claimed, and that journals are reluctant to permit the word otherwise.

Sourcing note: the current PDG scale-factor formula and rationale are firmly verifiable, as is the 1957 PDG origin and the statistical lineage to Birge (1932). The exact earliest *Review of Particle Properties* edition (likely 1969–1973) in which the scale-factor procedure first appears in print could not be confirmed from currently fetchable online archives and is flagged for primary-source follow-up.

C.6 Instrument grammar and traceability (L1, L9): 1887 → 1980 → 1993 → 1999

The uncertainty grammar and the traceability chain were the last layers to close, and they closed together. National metrology institutes anchored physical traceability: the Physikalisch-Technische Reichsanstalt (PTR), the world’s first NMI, was funded 28 March 1887; the UK’s NPL followed in 1900 and the US NBS (renamed NIST in 1988) in 1901. The International Organization of Legal Metrology (OIML) was created by convention in 1955. But the grammar for expressing measurement uncertainty was harmonized only in the 1977–1993 window: CIPM Recommendation INC-1 (1980) provided the foundational recommendation, leading to the Guide to the Expression of Uncertainty in Measurement (GUM), first published in 1993 (ISO/IEC Guide 98:1993; now JCGM 100:2008). The companion vocabulary (VIM) had its first edition in 1984 (now JCGM 200:2012). The international traceability/conformance chain was completed by ISO/IEC 17025 (first edition 1999, with lineage from ISO Guide 25, 1978) and the CIPM Mutual Recognition Arrangement, signed in Paris on 14 October 1999 by the directors of 38 NMIs and two international organizations. The felt crisis here was globalized trade and high-technology manufacturing requiring cross-border acceptance of calibration certificates.

The cost of a break in this chain at the engineering interface is illustrated by the loss of NASA’s Mars Climate Orbiter on 23 September 1999. The Mishap Investigation Board (Phase I Report, 10 November 1999, chaired by Arthur G. Stephenson) determined that the root cause was the failure to use metric units in a ground software file used in trajectory models — thruster performance data supplied in pound-force-seconds where the navigation software expected newton-seconds, a factor-of-4.45 unit-interface failure.

Appendix D: The score register

Table 1 is not maintained in this document. It is generated from `standardization-chart.v1.json`, a versioned register published under CC BY 4.0 at `nutri-collective.mealcoach.ai`. **The register is canonical; the HTML pages that display it** — the standardization chart and the source register at `/standardization-sources.html` — **are generated views, regenerated by script and never hand-edited.**

The register carries four things this paper cannot: (a) construct, the definition of what a score measures, reproduced as §2.2; (b) `rescore_log`, every score change since the 15 August 2026 recalibration with its reason and evidence; (c) superseded, the prior taxonomy retained so the migration remains auditable; and (d) per-layer rationales longer than a table cell allows. A companion register, `standardization-sources.v1.json`, carries 95 sources (as of 31 August 2026) with every DOI resolved against Crossref, and states in its own scope note that it sources the **claims** behind each cell and never the **scores**. No source is attached to any score, because nothing outside this project published these numbers.

The register also records the layer-threshold count at `functional_floor.layer_threshold`, recomputed from the scores on every build rather than asserted, and guarded by tests that fail the build if any future rescore lifts a nutrition layer to 60 while the text asserts that none does. The key was `functional_floor.result` before the terminology split made in §5 of this version; it was renamed at register revision 2, and the block records that earlier versions used the retired wording, so a reader diffing registers finds a correction rather than a silent rename.

Four guards run on every build. `test_chart_matches_the_published_table` prevents the site showing numbers the deposited paper does not. `test_material_rescores_are_on_the_record` prevents a score moving ≥ 10 points on an unchanged construct with no recorded reason and evidence. `test_scope_note_admits_open_disputes` prevents a disputed score being displayed without the dispute visible on the same page. `test_suite_cites_the_current_paper_version` prevents the validation protocol validating a superseded deposit.

Each register file carries its own licence block rather than relying on the site’s terms, so that a file travelling as a deposit supplement arrives with its terms attached; each states its scope honestly, including that the scores are first-party ordinal judgments attributable to no external party, that the source register reproduces no source’s text, that the adoption register is editorial and the organisations it names have not endorsed their entries, and that MI-Nutrition v0.1 is a Draft/RFC adopted by nobody.

Lifecycle. The register carries a lifecycle field: living while under review, hardened thereafter. While living, a score may change if it carries a `rescore_log` entry; once hardened, a score may change only with a version bump and an erratum. A test — `test_register_carries_the_lifecycle_field_the_paper_describes` — fails the build if this paragraph outruns the file it describes. The claim this paper makes for its numbers is not that they are right. It is that every one of them is derived, dated, and that every change to one is on the record.

Reviewers who wish to check a number should read the register rather than this appendix, because the register is what the tests run against.

Appendix E: Validation protocol

The framework ships a validation protocol specifying which claims about Table 1 can fail, a JSON schema for a scored layer pass, and a worked pass against an external object (the Malnutrition Care Score eCQM, CMS986). **The protocol has been specified and not yet run:** no inter-rater exercise has been conducted and the framework has no external validation. What is automated today is *conformance*, not *validity* — cases are checked against the schema, scores against the rubric’s band cut points, and the copied Table 1 column against the live register, so a correction to Table 1 cannot leave a stale number behind in a validation case. The suite is at `validation/`: `VALIDATION_SUITE.md` (the protocol), `nine-layer-case.schema.json` (the case schema), and `cases/` (scored layer passes, currently one real case plus a template). Its own status line reads *protocol (not yet run)*.

Appendix F: The adoption register

Alongside the score register the framework maintains an adoption register of 21 rows on a seven-stage ladder — not_started → proposed → under_discussion → piloting → endorsed → required → implemented — in three lanes: eight mature standards the framework binds to, eleven elements it proposes, and two reference entries. The separation is deliberate and the result is one-sided: of eight mature standards, seven are implemented and one is endorsed; of eleven proposed elements, **none has been endorsed or implemented by any external party**, and the three recorded as piloting are piloted by this project alone. That is recorded as piloting rather than implementation because self-adoption is the weakest available evidence of uptake, and a build test enforces it: a row naming this project as its own adopter cannot claim implementation.

The register is editorial and says so. It is not an official statistic and not a claim that adoption is occurring; each stage is asserted from dated evidence listed on its row, and a stage claim without a linked source fails the build rather than rendering as a quiet assertion. Corpus mention counts, where shown, are floors on use and never compliance rates — the same standard returns counts differing by more than an order of magnitude depending on the corpus and the field searched, so no count appears without its denominator. Citations of this paper are explicitly excluded from the register; only signals that change researcher or machine behaviour are tracked.

The register is included not because it shows adoption but because it shows its absence, at the layers where this paper argues the field is weakest — the proposed elements concentrate in L7, L8 and L9, which is the intended shape rather than an accident. A framework that scored its own uptake generously would fail the construct it applies to everyone else.

A naming caution. In the dietetics literature “MI” denotes motivational interviewing, and in clinical contexts myocardial infarction. A corpus scan for “MI-Nutrition” over 2016–2026 returns eight hits, all eight false positives of exactly this kind. The count is zero because all eight are noise, not because the query was never run. A standard intended to be measured by corpus scan must be named so that the scan finds it and finds only it; a rename is under consideration.

Author’s note on scope and integrity of claims

This working paper deliberately avoids several claims that circulate in adjacent literature but are not supported by primary sources: it does not attribute empirical results to protocol-only references; it does not treat the paper’s own coinages (population fence, value lane, prevention/treatment/reversal lanes, functional floor, MI-Nutrition, study value hypothesis) as established external concepts, nor attribute them to any external publication; it does not describe nutrition standardization as wholesale nonexistent; and it uses the NLEA (1990) and MIAME (2001) as its load-bearing analogies rather than more distant regulatory precedents. Physics and nutrition layer scores are presented throughout as calibrated ordinal expert judgments, not measurements. Where a claim rests on a document that was not read in full — EN 16104:2012, LanguaL — that is stated at the point of use and in Limitations.

¹ Thiel, P. & Masters, B. *Zero to One: Notes on Startups, or How to Build the Future*. Crown Business, 2014, p. 103. The passage is an aside in a book about startup strategy, not a scientific assessment, and the Harvard claim concerns undergraduate majors rather than the graduate nutrition faculty that does exist there. It is cited as a widely-read outside characterization of the field’s institutional standing, not as evidence for any score in Table 1.

² Local PubMed nutrition slice, 2,022,437 records, titles and abstracts only, queried 31 August 2026. Case-insensitive match on `standardi[sz]ation, harmoni[sz]ation, interoperab, reporting guideline, minimum information, controlled vocabular, FAIR (data|principle); deduplicated by record. Epidemiological rate adjustment is excluded explicitly — age.standardi[sz]. (direct|indirect) standardi[sz]ation, standardi[sz]ation of (rates|mortality|incidence)` — removing 92 records, because that sense of the word is unrelated to data standards. The comparison term is `standardi[sz]ed` (31,337 records), counted separately and never summed with the first set. **This is a floor on discussion, not a measure of practice:** title and abstract only, and full-text search of the same terms returns roughly 27× more; “standardization” has no controlled vocabulary, so a different term list yields a different floor. The sharpest values are in the tail — FAIR appears in 18 records, `minimum information` in 28, `controlled vocabular` in 69. The figure counts papers that *mention* standardization, not papers *about* it; §4 argues at length that a mention is not a practice.